

sg13g2_stdcell_hv — Generation of a Thick-Oxide (3.3 V) Standard Cell Library



IHP SG13G2 open PDK · derivation, all views, verification and physical sign-off

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1 Scope and result

The IHP SG13G2 open PDK ships no 3.3 V standard cells: its thick-oxide devices appear only in the sg13g2_io pad ring. sg13g2_stdcell_hv fills that gap — **all 84 cells** of sg13g2_stdcell rebuilt on sg13_hv_nmos / sg13_hv_pmos, same topology, pin names and pin order, named sg13g2_hv_* so both libraries coexist in one netlist.

Deliverable	Coverage	Status
SPICE netlist (spice/)	84 cells, 924 devices	verified against 3 independent views
CDL netlist (cdl/)	84 cells + 2 tie cells	LVS reference, all 68 drawn cells match
Verilog (verilog/)	84 modules	shared ihp_* UDPs, deliberately not duplicated
xschem symbols / schematics	84 + gallery sheet	netlist-equivalence proven
GDS layout (gds/)	68 cells (66 retargeted + 2 tie cells built here)	DRC clean, LVS clean
LEF abstracts (lef/)	68 macros + CoreSiteHV site	generated from the GDS, pin sets verified against CDL
Liberty NLDM (lib/)	66 cells, 600 timing tables	combinational + sequential; areas from layout

Physical sign-off, one fixed invocation of the PDK's own klayout decks — on the abutted two-row array and on the stricter **shared-rail array** (rows at the true 7.14 μm pitch, mixed and mirrored vertical neighbours — what a placed block actually produces):

Check	This library	IHP thin-oxide control (same harness)
DRC, cell rules	0	0
DRC, cell rules, shared-rail mixed array	0	—
DRC, chip-level metal density	7	6
LVS	68 / 68 match	fill cells fail identically without the documented relaxation

The density items are harness properties, not cell defects: metal density is a check on a *filled* die, and the one differing count (M1.j, min 35 % Metal1) reads 35.0 % over the padded array's bounding box against 37.1 % over the actual cell area (control: 45.2 %).

The library has also carried a real block: two variants of the `spi_slave` IP placed and routed through LibreLane/OpenROAD at 100 MHz pass the full IHP signoff deck — zero geometry violations (section 6).

2 The device transform

	thin oxide	thick oxide
device model	sg13_lv_nmos / sg13_lv_pmos	sg13_hv_nmos / sg13_hv_pmos
gate length	130 nm	450 nm (Gat.a3 minimum)
NMOS width	—	unchanged
PMOS width	—	× 2.40 , snapped per finger to the 5 nm grid
as/ad/ps/pd	—	recomputed from device geometry
supply	1.2 V	3.3 V

Deliberate long-channel devices (decap 1.0 μm , sighold 700 nm, dlygate4sd3_1 500 nm) keep their lengths; 914 devices moved to 450 nm. The antenna cell is unchanged apart from its name (junction diodes).

Why × 2.40. The thin-oxide library is sized for a centred switching threshold ($V_m/V_{DD} = 0.5046$ at $W_p/W_n = 1.5135$), not drive match. At 3.3 V / 450 nm the PMOS is far weaker (219 vs 533 $\mu\text{A}/\mu\text{m}$) and the same V_m needs $W_p/W_n = 3.63$, i.e. $K_p = 3.63/1.5135 = 2.40$ on every PMOS with every NMOS untouched — preserving each cell's internal stack ratios. Across the library's 0.3–17.9 μm width range the required ratio stays within $\pm 5\%$ of 2.40 (work/vm_sweep.py, work/ratio_check.py).

Parasitics follow the vendor deck's geometry formulas; before any generation they were validated by recomputing all 924 thin-oxide devices (worst relative error 3.75×10^{-4} , the vendor's four printed digits). The generator refuses to run if this validation fails.

Cost, measured on `inv_1` (`work/fo4.py`): input capacitance $2.67 \rightarrow 5.87$ fF (2.20×), FO4 delay $53.6 \rightarrow 142.4$ ps (2.66×). The thick-oxide NMOS delivers *more* current per micron at 3.3 V than the thin-oxide at 1.2 V (533 vs 391 $\mu\text{A}/\mu\text{m}$) — the loss is capacitance, not drive.

3 The views

All netlist views are re-emitted from one internal model by `work/gen_hv_lib.py`, so a device cannot silently diverge between views.

- **spice/** — 84 subckts, 924 PSP103 devices (OSDI/ngspice); widths synchronised to the drawn layout.
 - **cdl/** — the same subcircuits as M-cards with `*.PININFO`, the LVS reference; compared field-by-field against SPICE by `verify_sch.py`.
 - **verilog/** — 84 modules; deliberately no copy of `sg13g2_udp.v` (the `ihp_*` UDPs are shared — a second copy would collide).
 - **sym/, sch/** — 84 symbols (thin-oxide drawn geometry, thick-oxide netlist prefix, `$:SG13G2_HV_SCH` resolution) and 84 schematics plus a generated gallery sheet; three-view consistency proven on all 924 devices.
 - **gds/** — 68 cells: 66 by 1-D retarget (section 4), tie cells built by `work/gen_tie_cells.py`. Uniform $7.140\ \mu\text{m}$ row height (17 tracks), every width a $0.48\ \mu\text{m}$ site multiple, all 5 758 contact/via cuts exactly $0.16 \times 0.16\ \mu\text{m}$, every rail tap on the site-centred $0.48\ \mu\text{m}$ grid (`work/fix_rail_contacts.py`, section 6).
 - **lef/** — 68 macros + CoreSiteHV, generated from the GDS: pins from the pin datatypes labelled by contained text, OBS = drawn metal minus pin geometry, antenna values recomputed from the netlist (the thin-oxide numbers are wrong for 3.5× longer gates). Parsed back with `klayout` and pin sets verified against the CDL.
 - **lib/** — Liberty NLDM at 3.3 V / 25 °C typical (section 7).
 - **klayout/** — a cell-library registration macro: with the repository's `klayout/` directory on `KLAYOUT_PATH`, all 68 drawn cells appear in KLayout's Instance dialog as library `sg13g2_stdcell_hv`.
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4 Layout generation

Hand-drawn 2-D layout cannot be regenerated by placement, but it can be **1-D retargeted**: monotone piecewise-linear coordinate maps applied to every vertex preserve topology, so

connectivity survives by construction (`work/layout_retarget.py`). The engineering is in the breakpoints and the exceptions:

- **x-map**: gates widen $0.13 \rightarrow 0.45 \mu\text{m}$ about their centres; contacted poly pitch $0.48 \rightarrow 0.80 \mu\text{m}$. One accepted consequence: `dlygate4sd2_1` staggers two gates at overlapping x, so its PMOS lands at a legal $0.625 \mu\text{m}$, carried consistently into every netlist view.
- **y-map**: PMOS band $\times 2.40$ plus three channel inserts for the thick-oxide clearances (`NW.d1.dig` $0.215 \mu\text{m}$, `pSD.j1` $0.100 \mu\text{m}$, `pSD.i1` rail insert $0.430 \mu\text{m}$), derived library-wide so every cell keeps one row height.
- **PMOS diffusion** is re-emitted per connected slab piece with channel slabs frozen — the only scheme that keeps exact 5 nm-grid device widths without merging, collapsing or renaming devices (each alternative was tried and caught by LVS or `Act.*` rules).
- **Cut layers are translated, never scaled**; contacts are re-tiled inside the new `Activ` at $0.36 \mu\text{m}$ pitch (`Cnt.b1` array spacing), grouped by polygon.
- **ThickGateOx** is drawn on the cell boundary grown $0.27 \mu\text{m}$ in x and $0.42 \mu\text{m}$ in y — the y-margin proven by the N-row edge experiment (the `TGO.a` count stays 2 for any N and moves with the outer edge).
- **Five Metal1 re-routes** (`M1E_EDITS` in the generator) resolve the `M1.e` wide-metal gaps the retarget manufactures; each edit is verified in code (polygon count, `M1.a/b/c1`, 5 nm grid).
- **Netlist ↔ layout sync**: the drawn width is authoritative; `work/sync_netlist_widths.py` brings SPICE and CDL to the drawn geometry per *finger* and recomputes parasitics.
- **Site and tracks**: `TRACK_PAD` stretches the mid-cell dead zone to make exactly 17 horizontal $0.42 \mu\text{m}$ tracks; `pad_to_site` pads each width to the $0.48 \mu\text{m}$ site ($13.75 \mu\text{m}$ total across the library). Mean cell area is **2.87×** the thin-oxide library (median 2.83, range 1.89–4.21).

5 Functional verification

Suite	Coverage	Result
<code>verify_logic.py</code>	60 combinational cells, 452 vectors, both libraries in one deck	PASS
<code>verify_seq.py</code>	16 stateful cells, 400 clocked samples	PASS
<code>verify_sch.py</code>	84 schematics + CDL vs SPICE, 924 devices field-by-field	PASS

The thin-oxide library is the golden reference — the transform changed devices, never logic. The 12 high-impedance states of the tri-state cells are exercised; illegal set/reset

combinations are excluded rather than counted as passes; both suites were re-run on the final, shipped netlists.

6 Physical sign-off

Methodology. Cells are checked in abutted context, never standalone. Two harnesses: `work/make_drc_top.py` (every cell in its own column, mirrored second row at the padded bbox pitch) for cell-internal rules, and `work/make_shared_rail_rows.py` — rows at the true 7.14 μm pitch, orientations N/S/FS, rotated cell order per row, cells advanced by LEF width — for everything only a placed block exhibits. Measurement rules, each bought with a wrong conclusion: one fixed runner invocation (counts from different flag sets are not comparable), diagnosis from flat `klayout.db` rule replication (deep-mode marker coordinates mis-attribute cells), and the IHP thin-oxide library run through the identical harness as control. Pass/fail is parsed from the report database, never the exit code.

DRC. 781 raw violations after the first retarget were driven to zero cell-rule items (density-only residue, same class as the control). One defect class survived the original harness because it could not express it: the retarget left each cell's **rail-tap contacts** at cell-specific x, and in a placed block — where *different* cells share every rail — partially overlapping taps merged into 0.19–0.32 μm bars: ~19 000 Cnt/CntB markers on the first `spi_slave` signoff. `work/fix_rail_contacts.py` re-tiles every rail tap onto the site-centred 0.48 μm grid, which both the FS row flip and placement's x-mirror preserve, with in-code guards for implant polarity, enclosures, gate clearance, contact spacing and tap continuity (1 134 → 1 810 rail contacts). Full analysis with figures: the companion report *The Shared-Rail Contact Clash in sg13g2_stdcell_hv* (`shared_rail_contact_fix.pdf`, distributed alongside the library rather than inside it).

LVS. All 68 drawn cells match, re-run in full after the re-tiling. The four device-less `fill_*` cells need the runner's own `--ignore_top_ports_mismatch` relaxation — IHP's own fill cells fail the identical flow the same way. LVS caught five defect classes dimensional checks could not (merged channels, collapsed notch devices, mis-paired fingers, a dropped substrate tie, stale diode geometry), which is why the per-cell sweep runs after every layout change.

Block-level validation. The `spi_slave` IP, in two variants, was synthesised, placed and routed with this library through LibreLane/OpenROAD at 100 MHz on ~354 × 400 μm dies. Both close timing with zero setup/hold violations at the characterised corner, pass routing DRC, antenna and Netgen LVS in the flow, pass gate-level simulation of the routed netlist (all four SPI modes on the all-modes variant), and pass the **full IHP signoff deck** with zero geometry violations — only the 8 chip-level density/filler markers a filled die resolves at tapeout. The flow settings this takes (flip-flop mapping onto `sdffbbp_1`, excluded cells, Metal1 routing, rails-only fillers) are documented in the `spi_slave-ihp` repository.

7 Liberty characterisation

lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib is produced with **CharLib** against the shipped, layout-synchronised SPICE netlist on the PSP103/OSDI models — 25 839 combinational simulations plus a 2 962-task sequential run: 600 delay/slew tables over 66 cells (52 combinational, 9 flip-flops, 5 latches), none empty. Grids are the thin-oxide grids rescaled by the measured 2.66× delay / 2.20× capacitance ratios; boolean functions are translated from the thin-oxide Liberty and checked by truth-table equivalence. Stock CharLib needed five committed tool workarounds (OSDI shim, ngspice-shared backend, case-insensitive supply lookup, Liberty syntax post-pass) and — for sequential cells, which stock CharLib cannot characterise at all — custom clk→Q and setup/hold-bisection procedures with a c2q-degradation pass criterion (work/seq_delay_procedure.py).

The shipped file is verified **as data** by work/verify_lib.py: structure (600/600 tables populated), cross-view (all pins exist in the CDL), physical (every area equals the drawn boundary to 1 nm²), monotonicity along the load axis (1 931/1 932 series, one documented pessimistic waiver: xnor2_1 glitch latching), input capacitance against an independent both-rails measurement (6.44 vs 5.87 fF, 9.7 % — this check caught a 6.8–7.5× Miller defect in CharLib’s default procedure), and sequential arcs (14/14 cells: delay + setup + hold present, nothing pinned at search bounds). A hand-written transient at a table corner agrees to 0.3 %.

An independent characteriser, **lctime**, was run over eight cells on the same grids and models: in the region STA exercises (real loads, slews < 1 ns) the two agree to a median 2.9 % on delays and 0.0 % on transitions; every disagreement region traces to a stimulus convention, and a direct measurement sides with the shipped table (+0.15 % vs –19 % at the probe point). lctime’s input capacitance (+52 %) also corroborates validating Cin against a direct measurement, not a characteriser.

Excluded from the Liberty, by configuration and with stated reasons: the device-less cells, tiehi/tielo, the statetable clock gates, and the 6 tri-state cells (CharLib cannot express high-impedance outputs).

8 Verification summary

Check	Method	Scope	Result
Parasitic formulas	recompute vendor as / ad/ps/pd	all 924 thin-oxide devices	PASS, worst error 3.75×10 ^{−4}
Combinational logic	ngspice vs thin-oxide golden	60 cells, 452 vectors, 12 high-Z states	PASS
Sequential logic	ngspice, clocked walk from reset	16 stateful cells, 400 samples	PASS
Three-view consistency	symbols vs SPICE vs CDL	84 cells, 924 devices	PASS

Check	Method	Scope	Result
DRC, padded array	PDK deck, fixed invocation	68 drawn cells	0 cell rules ; 7 density items
DRC, shared-rail array	same deck, true-pitch mixed/mirrored rows	68 cells × 4 rows	0 cell rules ; density only
DRC control	identical harness on IHP sg13g2_stdcell	84 cells	0 cell rules, 6 density
LVS	PDK deck, per cell, after the rail re-tiling	68 cells vs CDL	68/68
LEF	klayout parse-back + pin sets vs CDL	68 macros	PASS, on-grid
P&R block validation	LibreLane spi_slave ×2 + full signoff deck	~354 × 400 μm, 100 MHz	all clean; signoff 0 geometry violations , 8 density markers
Liberty structure/ views/areas	verify_lib.py 1–3	66 cells	PASS, areas exact to 1 nm ²
Liberty monotonicity	verify_lib.py 4	1 932 delay series	1 931 + 1 documented waiver
Liberty Cin	vs both-rails reference	inv_1	6.44 vs 5.87 fF (9.7 %)
Liberty delay point-check	hand-written transient	table corner	0.3 %
Independent characteriser	lctime, 8 cells, 3 132 points	STA region	median 2.9 % / 0.0 %
Sequential arcs	verify_lib.py 6	14 flip-flops/latches	14/14 clean
Site geometry	boundary scan	all 68 cells	7.140 μm, widths on 0.48 μm site

9 Known limitations

- **16 cells have no layout** — eight D-flip-flop variants run NMOS Activ to the channel cut, eight others run PMOS Activ into the VDD rail; neither scales at a shared row height. All have full netlist, schematic, symbol and Verilog views. (tiehi/tielo, once on this list, were rebuilt by work/gen_tie_cells.py.)
- **P&R needs non-standard flow settings** — flip-flop mapping onto sdfbbp_1, an excluded-cell list, Metal1 routing for the off-grid pins, rails-only fillers (see the spi_slave-ihp flow configuration).
- **The 6 tri-state cells carry no Liberty timing**; they are simulation-verified.

- **One corner** (typical, 3.3 V, 25 °C); sequential leakage is a single-settled-state number; the delay cells' ratios shift with the 450 nm minimum; the decaps store less per unit area.
- **Not silicon-proven, and not independently reviewed.**

10 Reproducing the library

```
cd /foss/designs/sg13g2_stdcell_hv/work

python3 gen_hv_lib.py           # netlists, schematics, symbols, Verilog
python3 gen_gallery.py          # gallery sheet
python3 layout_retargt.py       # gds/ (66 cells; prints the 16 skips)
python3 gen_tie_cells.py        # + tiehi / tielo
python3 fix_rail_contacts.py    # rail taps onto the site-centred grid
python3 sync_netlist_widths.py  # SPICE + CDL follow the drawn geometry
python3 gen_lef.py              # lef/ (CoreSiteHV + 68 macros)
python3 make_drc_top.py         # padded two-row DRC context
python3 make_shared_rail_rows.py # shared-rail mixed-row DRC context

python3 verify_logic.py; python3 verify_seq.py; python3 verify_sch.py
./run_lvs.sh                    # per-cell LVS, 68/68
# DRC: PDK run_drc.py on drc/drc_top.gds and drc/shared_rail.gds

./run_charlib.sh ../lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib
# sequential cells via charlib_patched.py, then:
python3 merge_lib.py ../lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib <seq.lib>
python3 seq_leakage.py
python3 verify_lib.py           # gates the shipped Liberty as data
python3 lctime_compare.py       # independent characteriser cross-check
```

Every number in this report is reproducible from these scripts; the work/ directory and the README.md are the provenance of record.